

Functional Annotation of *hemC* (Q88RE5) — Porphobilinogen Deaminase / Hydroxymethylbilane Synthase in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

hemC (UniProt Q88RE5; ordered locus PP_0186) of *Pseudomonas putida* KT2440 encodes **porphobilinogen deaminase (PBGD)**, also called **hydroxymethylbilane synthase (HMBS)** or **pre-uroporphyrinogen synthase** (EC 2.5.1.61). It is the **third enzyme of the tetrapyrrole (heme) biosynthetic pathway**. The enzyme catalyzes the ordered, head-to-tail condensation of **four molecules of the monopyrrole porphobilinogen (PBG)** into the **linear tetrapyrrole 1-hydroxymethylbilane (pre-uroporphyrinogen)**, releasing four molecules of ammonia. Catalysis occurs on a **covalently bound dipyrromethane (DPM) cofactor** attached through a thioether bond to an invariant active-site cysteine. It is a **soluble, cytoplasmic** enzyme, and *hemC* is an **essential gene** whose product supplies the linear tetrapyrrole precursor for heme (and siroheme) biosynthesis.

Gene/Protein Identity Verification

- **Symbol vs. description:** The symbol *hemC* matches the UniProt description ("Porphobilinogen deaminase ... Hydroxymethylbilane synthase ... EC 2.5.1.61"). ✓
- **Organism:** *Pseudomonas putida* strain KT2440 (PSEPK); locus PP_0186. ✓
- **Family/domains:** InterPro annotations — HemC (IPR000860), Porphobilinogen_deaminase_N (IPR022417), Porphobilinogen_deaminase_C (IPR022418/IPR036803), Porphobilin_deaminase_cofac_BS (IPR022419) — are precisely the signatures of the **HMBS family** described in the structural and mechanistic literature below. ✓
- **No ambiguity:** *hemC* is a universally conserved heme-pathway gene; the literature cited concerns exactly this enzyme family. Because *P. putida* PBGD is highly conserved with the crystallographically characterized *E. coli*, human and *Arabidopsis* enzymes, mechanistic and structural inferences transfer directly.

2. Primary Function: Reaction and Substrate Specificity

Reaction (EC 2.5.1.61):



PBGD "catalyses the polymerization of four molecules of porphobilinogen to form the 1-hydroxymethylbilane, preuroporphyrinogen, a key intermediate in the biosynthesis of tetrapyrroles" (Louie *et al.*, 1996,

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HMBS "combines four porphobilinogen (PBG) molecules to form a linear tetrapyrrole, hydroxymethylbilane" (Sato *et al.*, 2021,

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- **Substrate:** the monopyrrole **porphobilinogen (PBG)**, itself produced by the preceding enzyme PBG synthase / ALA dehydratase (HemB). Each PBG bears an acetate, a propionate, and an aminomethyl group.
- **Product:** **1-hydroxymethylbilane (pre-uroporphyrinogen)**, an unstable linear tetrapyrrole.
- **Chemistry:** deamination coupled to C–C bond formation. During each of the four condensations an aminomethyl group is lost as ammonia and a methylene bridge links successive pyrrole units, extending the chain in a strictly ordered A-B-C-D fashion.
- **Substrate recognition:** a single binding pocket recognizes the PBG side chains. In the human enzyme these contacts involve Arg26/Ser28 (acetate), Arg173 (propionate), Gln34 (aminomethyl) and Asp99 (pyrrole nitrogen) (Sato *et al.*, 2021,

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the equivalent residues are conserved in bacterial orthologs including *P. putida*.

3. Catalytic Mechanism

The dipyrromethane (DPM) cofactor. Uniquely among enzymes, PBGD builds its product on its **own covalently attached DPM cofactor**, which acts as a **primer**. The cofactor "is linked covalently to the enzyme through a thioether bridge to the invariant Cys254" (*Arabidopsis* numbering; Roberts *et al.*, 2013,

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and "is covalently attached to a cysteine side-chain borne on a flexible loop of domain 3 ... [and] serves as a primer for the assembly of the tetrapyrrole product" (Louie *et al.*, 1996,

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The invariant catalytic cysteine corresponds to Cys242 in *E. coli* and Cys261 in human PBGD, with the equivalent conserved cysteine present in the *P. putida* sequence. Bacterial HemC directly carries this cofactor: in *Desulfovibrio vulgaris*, "PBG deaminase (HemC) was shown to contain the dipyrromethane cofactor" (Lobo *et al.*, 2009,

P 19267692

Single site, four steps. High-resolution complexes of holo-HMBS with the reaction intermediate carrying two bound PBG (the "ES2" state) show that "each of the four PBG molecules covalently binds to the cofactor (or oligopyrrole chain) consecutively, ultimately forming a hexapyrrole chain" at a **single substrate-binding site** (Sato *et al.*, 2021,

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Thus one active site sequentially elongates the chain from the DPM primer (2 pyrroles) up to a covalently tethered **hexapyrrole**, after which the four newly added pyrroles are released as the tetrapyrrole product, regenerating the DPM cofactor.

Key catalytic residues (mutagenesis evidence). - **Asp99** is catalytically essential: the D99G mutant is "an inactive holo-protein that exists as a complex with two substrate molecules covalently bound to the dipyrromethane cofactor" (Shoolingin-Jordan *et al.*, 2003,

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- **Arg149 and Arg173** are required for **DPM cofactor assembly**; R149Q and R173Q cannot reconstitute the holo-enzyme

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- **Arg167** is required for the catalytic cycle; R167Q accumulates stable enzyme–substrate intermediates

([P 12773194](#)).

Cofactor self-assembly. The DPM cofactor is generated by the apo-enzyme itself from pre-uroporphyrinogen; binding of the cofactor increases the thermal stability of the enzyme, and the structure "loosens" as the pyrrole chain elongates, priming product release (Bustad *et al.*, 2013,

[P 23815679](#)).

4. Structure and Structural Role

PBGD is a **monomeric, three-domain α/β protein**. In *E. coli*, "the polypeptide chain of PBGD is folded into three alpha/beta domains" (Louie *et al.*, 1996,

[P 8727319](#)):

domains 1 and 2 share a five-stranded mixed β -sheet topology and form an extensive **active-site cleft** at their interface, connected by two hinge segments; domain 3 (a three-stranded antiparallel sheet) carries the flexible cofactor loop. Inter-domain **hinge motion** accommodates the growing tetrapyrrole chain — mutation of the hinge residue His120 (human) to proline abolishes activity (Song *et al.*, 2009,

[P 18936296](#)).

The fold is conserved across the *E. coli* (1.76 Å), human (2.2 Å) and *Arabidopsis* (1.45 Å) structures, providing a reliable template for the *P. putida* enzyme.

Sequence-level confirmation for the *P. putida* protein (this study)

A pairwise alignment of **Q88RE5 (313 aa)** with the crystallographically defined *E. coli* PBGD (**P06983**, 313 aa) gives **66.6 % identity over the full length** — the two proteins are the same size and clearly orthologous. Critically, **every catalytically important residue is conserved at the identical aligned position** in the *P. putida* enzyme:

Residue (E. coli)	<i>P. putida</i> (Q88RE5)	Role
Cys242	Cys242	Thioether anchor of the dipyrromethane cofactor
Asp84 (= human Asp99)	Asp84	Key catalytic aspartate for deamination/condensation
Arg11, Arg131, Arg132, Arg149, Arg155	conserved (all Arg)	Substrate side-chain / cofactor binding, cofactor assembly
SSLRRQ cofactor-binding motif	present	Cofactor/oligopyrrole anchoring loop

This is direct sequence evidence — beyond family inference — that Q88RE5 has an **intact PBGD active site** and operates by the same DPM-cofactor mechanism as the structurally characterized enzymes.

5. Localization

PBGD/HemC is a **soluble cytoplasmic enzyme**. It carries no signal peptide, transmembrane segment, or membrane-anchor, consistent with the general biochemistry of the "common" (early) tetrapyrrole pathway, which operates in the cytosol up to the membrane-associated terminal steps (e.g., ferrochelatase HemH, which is membrane-associated — Guégan *et al.*, 2003,

P 12864856

The reaction — a soluble small-molecule polymerization producing a diffusible linear bilane — takes place in the cytoplasm, where its product is immediately channeled to the next (cytosolic) enzyme.

6. Pathway Context

hemC functions in the **common tetrapyrrole / heme biosynthesis pathway**, the branch point that supplies heme, siroheme, and (in relevant organisms) cobalamin and chlorophyll:

Glutamate → (C5 pathway: GltX/HemA/HemL) → 5-aminolevulinic acid (ALA)
 → (HemB, ALA dehydratase) → porphobilinogen (PBG)
 → (HemC, PBGD) → 1-hydroxymethylbilane ← THIS ENZYME
 → (HemD, uroporphyrinogen III synthase) → uroporphyrinogen III
 → ... → protoporphyrin IX → (HemH) → heme

The product of HemC is normally passed to **HemD (uroporphyrinogen III synthase)**, which cyclizes and rearranges the D-ring to form the physiological **uroporphyrinogen III** (Shoolingin-Jordan, 1995,

).

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In *P. putida* KT2440 this partner enzyme is encoded by a **dedicated, adjacent gene** — ***hemD* (Q88RE4, locus PP_0187, EC 4.2.1.75, 255 aa)** — **immediately downstream of *hemC* (PP_0186)**, forming a *hemC*–*hemD* gene pair. The presence of a separate HemD establishes that *P. putida* HemC is the **monofunctional** deaminase (the bifunctional PBGD/UROS arrangement is found only in organisms lacking *hemD*, e.g., *Leptospira* — Guégan *et al.*, 2003,

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and the genomic adjacency is consistent with hand-off/channeling of the labile bilane to HemD.

UniProt/HAMAP (rule MF_00260) annotation of Q88RE5 independently corroborates every functional assignment in this report: the catalytic activity "**4 porphobilinogen + H₂O = hydroxymethylbilane + 4 NH₄⁺**" (Rhea RHEA:13185; EC 2.5.1.61), a covalently bound **dipyrromethane cofactor** (ChEBI:60342; "Binds 1 dipyrromethane group covalently"), and the pathway assignment "**protoporphyrin-IX biosynthesis; coproporphyrinogen-III from 5-aminolevulinate: step 2/4.**" Without HemD, the labile 1-hydroxymethylbilane cyclizes non-enzymatically to the useless uroporphyrinogen I isomer — underscoring that HemC's role is the ordered assembly of the linear precursor, while ring closure/isomer selection is HemD's job. In bacteria the reconstituted HemC+HemD system converts PBG to uroporphyrinogen III in vitro (Raux *et al.*, 2003,

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and HemC activity governs flux of porphyrin intermediates in engineered pathways (Kim *et al.*, 2009,

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P 19473249

Lee *et al.*, 2013,

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Essentiality. *hemC* is essential for heme synthesis. A psychrophilic *hemC* was used to render *Francisella novicida* temperature-sensitive by substituting the essential gene (Pankowski, 2016,

P 27248501

), and a *hemC* mutation in *Salmonella* produces a growth-restricted small-colony variant that is compensated by gene amplification/second-site mutation (Pränting & Andersson, 2011,

P 21219453

). This predicts that *P. putida* PP_0186 is required for normal aerobic growth, since heme is the cofactor of its respiratory cytochromes and catalases.

7. Supported and Refuted Hypotheses

Hypothesis	Status	Basis
Q88RE5 = porphobilinogen deaminase (HMBS), EC 2.5.1.61	Supported	Domain architecture (IPR000860 etc.) + conserved HMBS family literature
Substrate is porphobilinogen; product is 1-hydroxymethylbilane	Supported	P 8727319 P 33600566
Catalysis uses a covalent DPM cofactor on an invariant Cys	Supported	P 23519422 P 8727319 P 19267692
A single active site performs all four condensations	Supported	P 33600566
Asp/Arg residues are catalytic / cofactor-assembly determinants	Supported	P 12773194 P 18936296
Cytoplasmic localization	Supported (inference)	No targeting signals; soluble-enzyme biochemistry
Enzyme is essential for heme synthesis / growth	Supported	P 27248501 P 21219453
<i>P. putida</i> HemC is bifunctional (also HemD activity)	Refuted	<i>P. putida</i> KT2440 encodes a separate, adjacent HemD (Q88RE4/PP_0187, EC 4.2.1.75); bifunctionality reported only for organisms lacking HemD (e.g., <i>Leptospira</i> ,) P 12864856
Catalytic residues (Cys242, Asp84, Arg cluster) conserved in Q88RE5	Supported	Pairwise alignment vs. <i>E. coli</i> P06983 (66.6% id): all catalytic residues conserved (this study)

8. Limitations and Future Directions

- **Direct experimental data for the *P. putida* KT2440 protein specifically are sparse;** functional and structural conclusions are inferred from its unambiguous membership in the highly conserved HMBS family and from crystallographic/mutagenesis work on the *E. coli*, human, *Arabidopsis* and other bacterial orthologs. The reaction, cofactor chemistry, and essentiality are so conserved that these inferences are robust.
- Localization is inferred from sequence features (no signal/TM segments) and general pathway biochemistry rather than a *P. putida*-specific fractionation experiment.
- **Future work:** (i) recombinant expression and kinetic characterization of PP_0186 (kcat/Km for PBG, DPM cofactor content); (ii) crystal/cryo-EM structure to confirm the conserved catalytic Asp/Arg constellation; (iii) targeted knockout/complementation in *P. putida* to confirm essentiality and any conditional heme-auxotrophy; (iv) test for possible HemC–HemD complex formation/substrate channeling.

References (PMIDs)

- 8727319 — Louie *et al.* 1996, *E. coli* PBGD structure at 1.76 Å.
- 7592565 — Shoolingin-Jordan 1995, PBGD & UROS structure/mechanism review.
- 33600566 — Sato *et al.* 2021, HMBS–substrate-analog complexes, single binding site.
- 23519422 — Roberts *et al.* 2013, *Arabidopsis* PBGD, DPM thioether to Cys254.
- 18936296 — Song *et al.* 2009, human PBGD structure; hinge residue.
- 12773194 — Shoolingin-Jordan *et al.* 2003, catalytic Asp99/Arg mutagenesis.
- 23815679 — Bustad *et al.* 2013, cofactor/stability during elongation.
- 19267692 — Lobo *et al.* 2009, *Desulfovibrio* HemC contains DPM cofactor.
- 27248501 — Pankowski 2016, *hemC* essential gene (temperature-sensitivity).
- 21219453 — Pr nting & Andersson 2011, *hemC* mutation growth restriction (*Salmonella*).
- 12408752 — Raux *et al.* 2003, HemC+HemD in vitro uroporphyrinogen III synthesis.
- 19473249 / 23648857 — HemC controls porphyrin flux in engineered *E. coli*.

- 12864856 — Guégan *et al.* 2003, *Leptospira* heme genes; bifunctional HemC only where HemD absent.

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